

034IN - FONDAMENTI DI AUTOMATICA - FUNDAMENTALS OF AUTOMATIC CONTROL A.Y. 2024-2025

The Laplace Transform: Definition, Properties, Application Examples

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What is a transform?

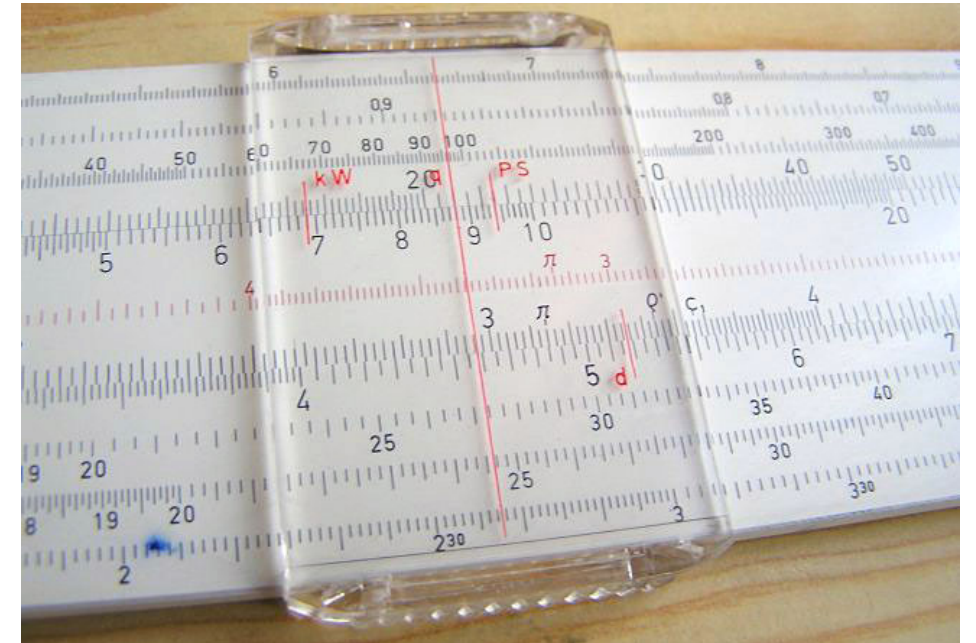
- A **mapping** of a mathematical function from one domain to another.
- A change in **perspective**, not a change of the function.

Why use transforms?

- Some mathematical **problems** are **difficult to solve** in their natural domain.
- Transform to and solve in a **new domain**, where **the problem is simplified**.
- Solved the problem in the new domain, transform the solution back into the original domain.
- Trade off the extra effort of transforming/inverse-transforming for simplification of the solution procedure.

A Simple Transform Example – Slide Rule

- Slide rules make use of a logarithmic transform.
- Multiplication/division of large numbers is difficult:
 - **Transform** the numbers to the logarithmic domain
 - Add/subtract (**easy**) in the log domain to multiply/divide (difficult) in the linear domain
 - Apply the **inverse transform** to get back to the original domain
- Extra effort is required, but **the problem is simplified!**



The Laplace Transform

Motivation and Definition

- An **integral transform** mapping functions from the **time domain** to the **Laplace domain** or s-domain.

$$g(t) \xleftrightarrow{\mathcal{L}} G(s)$$

- Time-domain functions are functions of time, t
- Laplace-domain functions are functions of s
- s is a complex variable $s = \sigma + j\omega$

$$\sigma = \operatorname{Re}(s) = \Re s$$

$$\omega = \operatorname{Im}(s) = \Im s$$

Laplace Transform - Motivation

- We'll use Laplace transforms **to solve differential equations**.
 - **Differential equations** in the **time domain**: difficult to solve.
 - Apply the Laplace transform: **transform to the s-domain**.
 - Differential equations become **algebraic equations**: easy to solve!
 - **Transform** the s-domain solution **back to the time domain**.
- Transforming back and forth requires extra effort, but the solution is greatly simplified!

Causal Time-Domain Functions

- We are interested in functions in the time domain defined for $t \geq 0$, assuming they are null for all negative times, i.e.

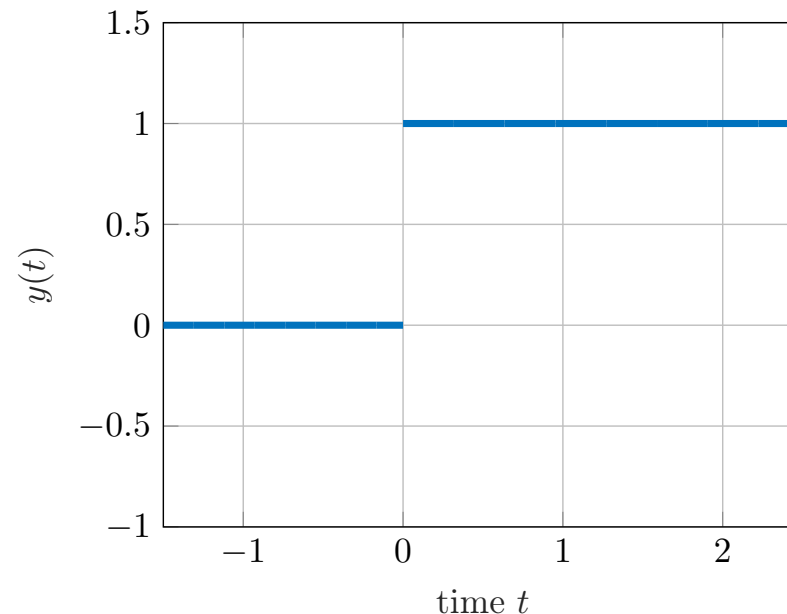
$$f(t) \equiv 0, \quad \forall t < 0$$

- Time domain functions that have this property are called **causal functions**.

Examples

- The **unit step function** or Heaviside step function

$$1(t) := \begin{cases} 0, & \forall t < 0 \\ 1, & \forall t \geq 0 \end{cases}$$



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Causal Time-Domain Functions: Notation

- We'll use the Heaviside function to **emphasise** that we are considering a causal function

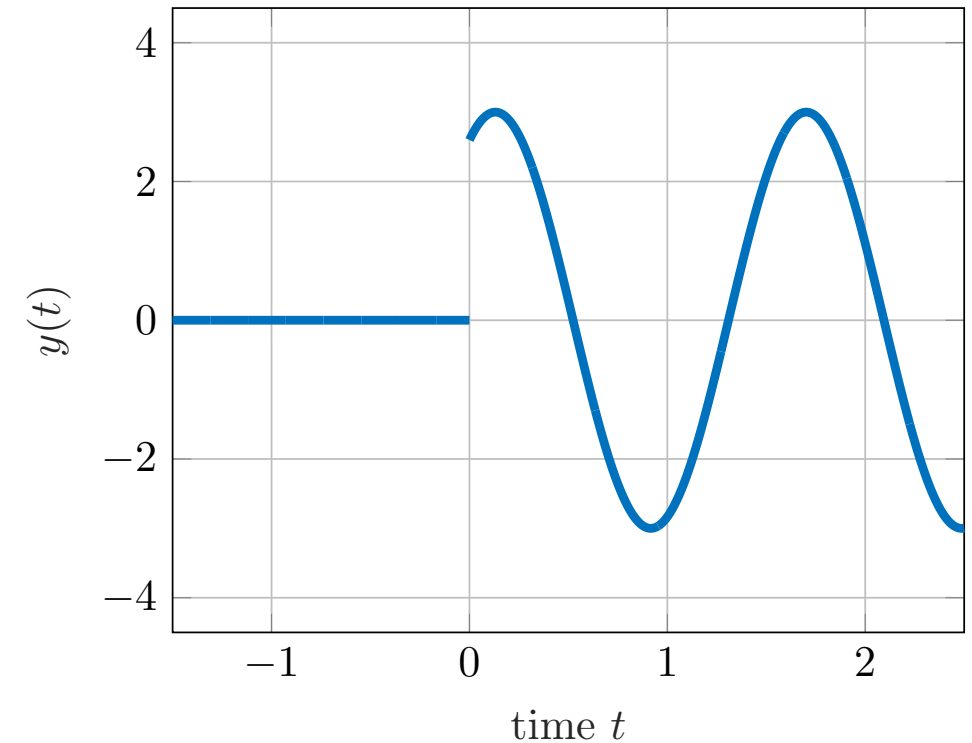
$$f(t) = \left[3 \sin \left(4t + \frac{\pi}{3} \right) \right] \cdot 1(t)$$

$$= \begin{cases} 0, & \forall t < 0 \\ 3 \sin \left(4t + \frac{\pi}{3} \right), & \forall t \geq 0 \end{cases}$$



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Laplace Transform - The Unilateral or One-sided Transform

- Given a **causal function** $f(t)$, the Laplace transform of $f(t)$ is the function $F(s) = \mathcal{L}(f(t))$ defined by

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

for those $s \in \mathbb{C}$ for which the integral makes sense.

- F is a complex-valued function of complex numbers
- s is called the *(complex) frequency variable*, with units sec^{-1} ; t is called the *time variable* (in sec); st is unit-less
- Lower limit of integration is $t = 0$: the transform is called the **unilateral** or **one-sided Laplace transform** (Laplace transform for short).

Laplace Transform - Notation Convention

$$F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

- **common notation convention:** lower case letter denotes the time-domain function $f(t)$; capital letter denotes its Laplace transform $F(s)$

- for example

$$U(s) = \mathcal{L}(u(t)) \qquad V_{\text{in}}(s) = \mathcal{L}(v_{\text{in}}(t))$$

Laplace Transform - Existence

$$f(t) \quad \overset{?}{\longleftrightarrow} \quad F(s) = \int_0^{\infty} f(t) e^{-st} dt$$

Which class or classes of functions possess a Laplace transform?

- **Fact 1:** $f(t)$ is **continuous** for $t = 0$; if not, $f(t)$ must contain **no impulses** at $t = 0$.
- **Fact 2:** any continuous, bounded, causal function $f(t)$ admits Laplace transform.
- **Fact 3:** any **piecewise continuous**, **causal** function $f(t)$ admits Laplace transform.

Piecewise continuous function $\longleftrightarrow$ jump discontinuities and no vertical asymptotes

Laplace Transform - Example: Exponential Function

- Let's find the Laplace transform of the causal function $f(t) = e^t \cdot 1(t)$

$$F(s) = \int_0^{\infty} e^t e^{-st} dt = \int_0^{\infty} e^{(1-s)t} dt = \frac{1}{1-s} e^{(1-s)t} \Big|_0^{\infty} = \frac{1}{s-1}$$

provided we can say $e^{(1-s)t} \rightarrow 0$ as $t \rightarrow \infty$, which is true for $\text{Re}(s) > 1$

$$\left| e^{(1-s)t} \right| = \left| e^{-j\omega t} \right| \cdot \left| e^{(1-\sigma)t} \right| = e^{(1-\sigma)t}$$

- The integral defining F makes sense for all $s \in \mathbb{C}$ with $\text{Re}(s) > 1$ (the 'region of convergence' of F)
- The resulting formula for F makes sense for all $s \in \mathbb{C}$, except $s = 1$

Laplace Transform - Example: Unit Step Function

- Let's find the Laplace transform of the causal function $f(t) = 1(t)$

$$F(s) = \int_0^{\infty} 1(t) e^{-st} dt = \int_0^{\infty} e^{-st} dt = -\frac{1}{s} e^{-st} \Big|_0^{\infty} = \frac{1}{s}$$

- The integral defining F makes sense for all $s \in \mathbb{C}$ with $\text{Re}(s) > 0$ (the 'region of convergence' of F)
- The resulting formula for F makes sense for all $s \in \mathbb{C}$, except $s = 0$
- We'll ignore these (sometimes important) details and just write $\mathcal{L}(1(t)) = \frac{1}{s}$

Laplace Transform - Properties: Linearity



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- The Laplace transform is **linear**

- If $f(t)$ and $g(t)$ are any causal functions, for which the Laplace transform exists, and a is any scalar, we have

$$\mathcal{L}(a f(t)) = a F(s) \quad \mathcal{L}(f(t) + g(t)) = F(s) + G(s)$$

i.e., homogeneity and superposition hold.

Example

$$\mathcal{L}(3 \cdot 1(t) - 2e^t \cdot 1(t)) = 3 \cdot \frac{1}{s} - 2 \cdot \frac{1}{s-1} = \frac{s-3}{s(s-1)}$$

Laplace Transform - Example: Trigonometric Functions

- Let's find the Laplace transform of the causal function $f(t) = [\cos(\omega t)] \cdot 1(t)$
- First step: express $f(t)$ as $f(t) = \left[\frac{1}{2}e^{j\omega t} + \frac{1}{2}e^{-j\omega t} \right] \cdot 1(t)$
- Second step: according to the *linearity* property

$$\begin{aligned} F(s) = \mathcal{L}(f(t)) &= \frac{1}{2} \mathcal{L}(e^{j\omega t} \cdot 1(t)) + \frac{1}{2} \mathcal{L}(e^{-j\omega t} \cdot 1(t)) \\ &= \frac{1}{2} \int_0^{\infty} e^{(-s+j\omega)t} dt + \frac{1}{2} \int_0^{\infty} e^{(-s-j\omega)t} dt \\ &= \frac{1}{2} \frac{1}{s-j\omega} + \frac{1}{2} \frac{1}{s+j\omega} = \frac{s}{s^2 + \omega^2} \end{aligned}$$

Laplace Transform - Example: Trigonometric Functions

$$f(t) = [\cos(\omega t)] \cdot 1(t) \quad \Longleftrightarrow \quad F(s) = \mathcal{L}(f(t)) = \frac{s}{s^2 + \omega^2}$$

- How do you do this in Matlab?

```
clear variables  
syms t s
```


Laplace Transform - One-to-One Property

- The Laplace transform is one-to-one:

$$f(t) = g(t) \iff \mathcal{L}(f(t)) = \mathcal{L}(g(t))$$

- $F(s)$ determines $f(t)$
- **Inverse Laplace transform** is well defined

Example (previous page)

$$\begin{aligned}\mathcal{L}^{-1} \left[\frac{s-3}{s(s-1)} \right] &= \mathcal{L}^{-1} \left[\frac{3}{s} - \frac{2}{s-1} \right] = \\ &= \mathcal{L}^{-1} \left[\frac{3}{s} \right] - \mathcal{L}^{-1} \left[\frac{2}{s-1} \right] = 3 \cdot 1(t) - 2e^t \cdot 1(t)\end{aligned}$$

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Laplace Transform - Transform of a Derivative

- If the function $f(t)$ is continuous at $t = 0$, then $\mathcal{L} \left(\dot{f}(t) \right) = s F(s) - f(0)$
- **time-domain differentiation** becomes **multiplication by frequency variable** s (as with phasors)
- plus a term that includes initial condition, i.e. $-f(0)$

Example

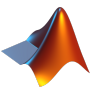
$$f(t) = 2e^t \cdot 1(t) \implies \dot{f}(t) = 2e^t \cdot 1(t)$$

$$\mathcal{L} \left(\dot{f}(t) \right) = s \left(\frac{2}{s-1} \right) - f(0) = \frac{2s}{s-1} - 2 = \frac{2}{s-1}$$



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$$\mathcal{L}(\dot{f}) = sF(s) - f(0)$$

- higher order derivatives: applying derivative formula twice yields

$$\begin{aligned}\mathcal{L}(\ddot{f}) &= s \mathcal{L}(\dot{f}) - \dot{f}(0) \\ &= s (s F(s) - f(0)) - \dot{f}(0) \\ &= s^2 F(s) - s f(0) - \dot{f}(0)\end{aligned}$$

- a similar formula holds for $\mathcal{L}(f^{(k)}) = s^k F(s) - \sum_{n=0}^{k-1} s^{k-n-1} f^{(n)}(0)$

Laplace Transform - Transform of an Integral

- Let g be the running integral of a causal function f

$$g(t) = \int_0^t f(\tau) d\tau$$

- then

$$G(s) = \frac{1}{s} F(s)$$

time-domain integral becomes division by frequency variable s

Laplace Transform - Transform of an Integral (cont.)

$$g(t) = \int_0^t f(\tau) d\tau \iff G(s) = \frac{1}{s} F(s)$$

Example

$$f(t) = 1(t) \iff F(s) = \frac{1}{s}$$

$$g(t) = \int_0^t 1(\tau) d\tau = t, \quad t \geq 0 \iff G(s) = \frac{1}{s} F(s) = \frac{1}{s^2}$$

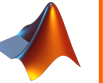


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Laplace Transform - Multiplication by t

- Let $f(t)$ be a causal function and define $g(t) = t \cdot f(t)$

- Then we have $G(s) = -\frac{d}{ds} F(s)$

- In general $g(t) = t^k \cdot f(t)$

$$G(s) = \left(-\frac{d}{ds}\right)^k F(s) = \underbrace{-\frac{d}{ds} \left[-\frac{d}{ds} \left[-\frac{d}{ds} \left[\cdots -\frac{d}{ds} F(s) \right] \right] \right]}_{k \text{ times}}$$

Laplace Transform - Multiplication by t (cont.)

■ Examples

$$f(t) = e^t \cdot 1(t), \quad g(t) = t e^t \cdot 1(t)$$

$$\mathcal{L}(g(t)) = -\frac{d}{ds} \left(\frac{1}{s-1} \right) = \frac{1}{(s-1)^2}$$

$$f(t) = t e^t \cdot 1(t), \quad g(t) = t^2 e^t \cdot 1(t)$$

$$\mathcal{L}(g(t)) = -\frac{d}{ds} \left(\frac{1}{(s-1)^2} \right) = \frac{2}{(s-1)^3}$$



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Laplace Transform - Multiplication by t (cont.)

- **Examples**

- In general

$$f(t) = e^t \cdot 1(t), \quad g(t) = t^k e^t \cdot 1(t)$$

$$\mathcal{L}(g(t)) = \mathcal{L}(t^k e^t \cdot 1(t)) = \frac{(k-1)!}{(s-1)^{k+1}}$$

Laplace Transform - Time Scaling

- Let $f(t)$ be a causal function, admitting Laplace transform, and let $g(t)$ be defined as

$$g(t) = f(at), \quad a > 0$$

then

$$G(s) = \frac{1}{a} F\left(\frac{s}{a}\right)$$

Remark: this makes sense! If time is scaled by a , then frequency

s is scaled by $\frac{1}{a}$

Laplace Transform - Time Scaling (cont.)

$$g(t) = f(at), \quad a > 0 \iff G(s) = \frac{1}{a} F\left(\frac{s}{a}\right)$$

Example

starting from $\mathcal{L}(e^t \cdot 1(t)) = \frac{1}{s-1}$



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we obtain

$$\mathcal{L}(e^{at} \cdot 1(t)) = \left(\frac{1}{a}\right) \frac{1}{\left(\frac{s}{a}\right) - 1} = \frac{1}{s-a}, \quad a > 0$$

- Let $f(t)$ be a causal function, with Laplace transform $F(s)$, and let $g(t)$ be defined as (a is a scalar value)

$$g(t) = e^{a t} f(t)$$

then

$$G(s) = F(s - a)$$

Remark: a frequency shift corresponds to a multiplication with an exponential term in the time domain!

Laplace Transform - Exponential Scaling (cont.)

Example: given $\mathcal{L}[(\cos(\omega t)) \cdot 1(t)] = \frac{s}{s^2 + \omega^2}$

then

$$\begin{aligned}\mathcal{L}[(e^{-t} \cos(t)) \cdot 1(t)] &= \frac{(s + 1)}{(s + 1)^2 + 1} \\ &= \frac{s + 1}{s^2 + 2s + 2}\end{aligned}$$

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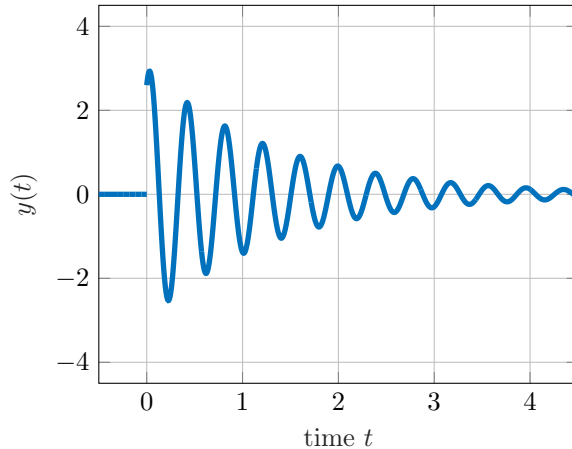


Laplace Transform - Time Shifting (Time Delay)

- Let $f(t)$ be a causal function, with Laplace transform $F(s)$, and $T > 0$
- Define $g(t)$ as

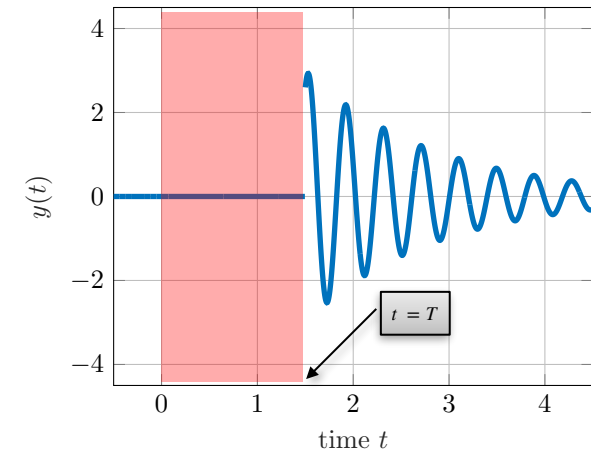
$$g(t) = \begin{cases} 0 & 0 \leq t < T \\ f(t - T) & t \geq T \end{cases} \longleftrightarrow g(t) = f(t - T) \cdot 1(t - T)$$

i.e., g is f , delayed by T seconds and 'zero-padded' up to T



$$f(t) = \left[3e^{-0.75t} \sin \left(16t + \frac{\pi}{3} \right) \right] \cdot 1(t)$$

$$g(t) = f(t - 1.5) \cdot 1(t - 1.5)$$



Laplace Transform - Time Shifting (Time Delay) (cont.)

- Let $f(t)$ be a causal function, with Laplace transform $F(s)$, and $T > 0$
- Define $g(t)$ as

$$g(t) = f(t - T) \cdot 1(t - T)$$

then

$$G(s) = e^{-sT} F(s)$$

Remark: a time shift corresponds to a multiplication with an exponential term in the frequency domain!

Laplace Transform - Time Shifting (Time Delay) (cont.)

Example: let's find the Laplace transform of a rectangular pulse signal

$$f(t) = \begin{cases} 1 & a \geq t \geq b \\ 0 & \text{otherwise} \end{cases}$$

where $0 < a < b$

Write $f(t)$ as a unit step function delayed by a seconds, minus a unit step function delayed by b seconds

$$\begin{aligned} f(t) = 1(t - a) - 1(t - b) &\longleftrightarrow F(s) = \mathcal{L}(1(t - a)) - \mathcal{L}(1(t - b)) \\ &= \frac{e^{-a s} - e^{-b s}}{s} \end{aligned}$$

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Laplace Transform - Convolution (in the Time Domain)

- The **convolution** of causal functions f and g is the causal function h

$$h := f * g \longleftrightarrow h(t) = \int_0^t f(\tau) g(t - \tau) d\tau$$

- **Note:** same as $h(t) = \int_0^t f(t - \tau) g(\tau) d\tau$, i.e. $f * g = g * f$

$$H(s) = \mathcal{L}(f * g) = F(s) G(s)$$

Remark: Laplace transform turns convolution in time domain into multiplication in the frequency domain!

Laplace Transform of Common Functions

$f(t)$	$F(s)$
$1(t)$	$\frac{1}{s}$
$t \cdot 1(t)$	$\frac{1}{s^2}$
$t^2 \cdot 1(t)$	$\frac{2}{s^3}$
$e^{at} \cdot 1(t)$	$\frac{1}{s - a}$
$te^{at} \cdot 1(t)$	$\frac{1}{(s - a)^2}$
$[\sin(\omega t)] \cdot 1(t)$	$\frac{\omega}{s^2 + \omega^2}$

$f(t)$	$F(s)$
$[\cos(\omega t)] \cdot 1(t)$	$\frac{s}{s^2 + \omega^2}$
$[t \sin(\omega t)] \cdot 1(t)$	$\frac{2\omega s}{(s^2 + \omega^2)^2}$
$[t \cos(\omega t)] \cdot 1(t)$	$\frac{s^2 - \omega^2}{(s^2 + \omega^2)^2}$
$[e^{\sigma t} \sin(\omega t)] \cdot 1(t)$	$\frac{\omega}{(s - \sigma)^2 + \omega^2}$
$[e^{\sigma t} \cos(\omega t)] \cdot 1(t)$	$\frac{s - \sigma}{(s - \sigma)^2 + \omega^2}$
$[t e^{\sigma t} \sin(\omega t)] \cdot 1(t)$	$\frac{2\omega(s - \sigma)}{[(s - \sigma)^2 + \omega^2]^2}$
$[t e^{\sigma t} \cos(\omega t)] \cdot 1(t)$	$\frac{(s - \sigma)^2 - \omega^2}{[(s - \sigma)^2 + \omega^2]^2}$

Inverse Laplace Transform

- In principle we can recover $f(t)$ from $F(s)$ via

$$f(t) = \frac{1}{2\pi j} \int_{\sigma - j\infty}^{\sigma + j\infty} F(s) e^{st} ds$$

where σ is large enough that $F(s)$ is defined for $\text{Re}(s) \geq \sigma$

- Surprisingly, this formula isn't really useful!



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Inverse Laplace Transform (cont.)

- The **functions** in the **time domain** $f(t)$, with which we will deal, will always be **causal, piecewise continuous**, and they contain **no impulses at the time** $t = 0$
- Thus, the corresponding **Laplace transforms** $F(s)$ will **always** be **rational polynomial functions**, i.e., the quotient of two polynomials

$$F(s) = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0} = \frac{N(s)}{D(s)} \quad \begin{array}{l} \forall j \quad b_j \in \mathbb{R} \\ \forall i \quad a_i \in \mathbb{R} \end{array}$$

- Furthermore, the degree of the numerator polynomial will be less than that of the denominator polynomial:

$$\deg(N(s)) = m < n = \deg(D(s))$$

Laplace Transform: Notation - Zeros and Poles

- Given a Laplace transform $F(s)$ as a rational polynomial function, taking in evidence the roots of both the polynomials, we define

$$F(s) = \frac{N(s)}{D(s)} = K \frac{(s - z_1)^{m_1} (s - z_2)^{m_2} \dots (s - z_r)^{m_r}}{(s - p_1)^{n_1} (s - p_2)^{n_2} \dots (s - p_q)^{n_q}}$$

zeros: the roots of the polynomial at the numerator $N(s)$

$$z_1, z_2, \dots, z_r \in \mathbb{C}, \quad \sum_{j=1}^r m_j = m = \deg(N(s))$$

poles: the roots of the polynomial at the denominator $D(s)$

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$$p_1, p_2, \dots, p_q \in \mathbb{C}, \quad \sum_{i=1}^q n_i = n = \deg(D(s))$$

Inverse Laplace Transform (cont.)

- Starting from a Laplace transform $F(s)$ that is a **strictly proper rational function**, it is always possible to express the function as a **sum of partial fractions**:

$$\begin{aligned} F(s) = & \frac{C_{1,1}}{(s - p_1)} + \dots + \frac{C_{1,n_1}}{(s - p_1)^{n_1}} + \dots \\ & \dots + \frac{C_{2,1}}{(s - p_2)} + \dots + \frac{C_{2,n_2}}{(s - p_2)^{n_2}} + \dots \\ & \dots + \frac{C_{q,1}}{(s - p_q)} + \dots + \frac{C_{q,n_q}}{(s - p_q)^{n_q}} \end{aligned}$$

$$C_{i,j} \in \mathbb{C}, \quad i = 1, 2, \dots, q, \quad j = 1, 2, \dots, n_i$$

Inverse Laplace Transform (cont.)

- Owing to the linearity of the inverse Laplace transform, the time domain function we are looking for will be the linear combination of elementary exponential functions:

$$\mathcal{L}^{-1}[F(s)] = \sum_{i=1}^q \sum_{j=1}^{n_i} \mathcal{L}^{-1} \left[\frac{C_{i,j}}{(s - p_i)^j} \right]$$

$$\mathcal{L}^{-1}[F(s)] = \sum_{i=1}^q \sum_{j=1}^{n_i} \frac{C_{i,j}}{(j-1)!} t^{(j-1)} e^{p_i t} \mathbf{1}(t)$$



Inverse Laplace Transform (cont.)

- How to calculate the $C_{i,j}$ coefficients?
- There is a not very easy formula:

$$C_{i,j} = \frac{1}{(n_i - j)!} \lim_{s \rightarrow p_i} \left\{ \frac{d^{(n_i - j)}}{d s^{(n_i - j)}} [F(s) (s - p_i)^{n_i}] \right\}$$

- We can exploit the **identity theorem for polynomials**.
- We will see both techniques in action by solving a linear differential equation using the Laplace transform.

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Example: Solution of a Linear Differential Equation

- We want to solve the ODE equation:

$$\frac{d^2 y(t)}{dt^2} + 3 \frac{dy(t)}{dt} + 2y(t) = [1 + 3t] \cdot 1(t)$$

with initial conditions $y(0) = 1 \quad \left. \frac{dy}{dt} \right|_{t=0} = 0$

- Applying the properties analysed so far, we obtain

$$\left\{ s^2 Y(s) - s y(0) - \dot{y}(0) \right\} + \{ 3 [s Y(s) - y(0)] \} + \dots$$
$$\dots + 2 Y(s) = \frac{1}{s} + \frac{3}{s^2}$$

Example: Solution of a Linear Differential Equation (cont.)

- Substituting the values of the initial conditions gives:

$$\left\{s^2 Y(s) - s y(0) - \dot{y}(0)\right\} + \{3[s Y(s) - y(0)]\} + \dots$$
$$\dots + 2 Y(s) = \frac{1}{s} + \frac{3}{s^2}$$

$$\left\{s^2 Y(s) - s\right\} + \{3[s Y(s) - 1]\} + 2 Y(s) = \frac{1}{s} + \frac{3}{s^2}$$

$$(s^2 + 3s + 2)Y(s) - s - 3 = \frac{1}{s} + \frac{3}{s^2}$$

Example: Solution of a Linear Differential Equation (cont.)

- We have obtained an algebraic equation that has $Y(s)$ as its solution.

$$Y(s) = \frac{s+3}{s^2+3s+2} + \frac{1}{s^2+3s+2} \left(\frac{1}{s} + \frac{3}{s^2} \right)$$

- The first term of $Y(s)$ depends on the initial conditions (the so-called **free trajectory**), and the second term depends on the input $u(t)$ (the so-called **forced trajectory**)

$$Y_l(s) = \frac{s+3}{s^2+3s+2}$$

$$Y_f(s) = \frac{s+3}{s^2(s^2+3s+2)}$$

Example: Solution of a Linear Differential Equation (cont.)

$$Y_l(s) = \frac{s+3}{s^2+3s+2} = \frac{C_1}{s+1} + \frac{C_2}{s+2}$$

- **First approach:** using the formula
$$C_{i,j} = \frac{1}{(n_i-j)!} \lim_{s \rightarrow p_i} \left\{ \frac{d^{(n_i-j)}}{ds^{(n_i-j)}} [F(s) (s-p_i)^{n_i}] \right\}$$

$$C_1 = \lim_{s \rightarrow -1} \frac{s+3}{(s+1)(s+2)} (s+1) = 2$$

$$C_2 = \lim_{s \rightarrow -2} \frac{s+3}{(s+1)(s+2)} (s+2) = -1$$

$$Y_l(s) = \frac{2}{s+1} - \frac{1}{s+2}$$

Example: Solution of a Linear Differential Equation (cont.)

$$Y_l(s) = \frac{s+3}{s^2+3s+2} = \frac{C_1}{s+1} + \frac{C_2}{s+2}$$

- **Second approach:** exploiting the **identity theorem for polynomials**.

$$\frac{s+3}{(s+1)(s+2)} = \frac{s(C_1+C_2) + (2C_1+C_2)}{(s+1)(s+2)}$$

$$Y_l(s) = \frac{2}{s+1} - \frac{1}{s+2}$$

$$\begin{cases} C_1 + C_2 = 1 \\ 2C_1 + C_2 = 3 \end{cases} \implies \begin{cases} C_1 = 2 \\ C_2 = -1 \end{cases}$$

Example: Solution of a Linear Differential Equation (cont.)

$$Y_l(s) = \frac{2}{s+1} - \frac{1}{s+2}$$

$$\mathcal{L}^{-1} \{Y_l(s)\} = y_l(t) = [2e^{-t} - e^{-2t}] \cdot 1(t)$$

Example: Solution of a Linear Differential Equation (cont.)

$$Y_f(s) = \frac{s+3}{s^2(s^2+3s+2)} = \frac{C_{1,1}}{s+1} + \frac{C_{2,1}}{s+2} + \frac{C_{3,1}}{s} + \frac{C_{3,2}}{s^2}$$

- **First approach:** using the formula $C_{i,j} = \frac{1}{(n_i-j)!} \lim_{s \rightarrow p_i} \left\{ \frac{d^{(n_i-j)}}{ds^{(n_i-j)}} [F(s)(s-p_i)^{n_i}] \right\}$

$$C_{1,1} = \left. \frac{(s+3)}{(s^2)(s+2)} \right|_{s=-1} = 2$$

$$C_{3,1} = \left. \frac{d}{ds} \frac{(s+3)}{(s+1)(s+2)} \right|_{s=0} = -\frac{7}{4}$$

$$C_{2,1} = \left. \frac{(s+3)}{(s^2)(s+1)} \right|_{s=-2} = -\frac{1}{4}$$

$$C_{3,2} = \left. \frac{(s+3)}{(s+1)(s+2)} \right|_{s=0} = \frac{3}{2}$$

Example: Solution of a Linear Differential Equation (cont.)

$$Y_f(s) = \frac{s+3}{s^2(s^2+3s+2)} = \frac{C_{1,1}}{s+1} + \frac{C_{2,1}}{s+2} + \frac{C_{3,1}}{s} + \frac{C_{3,2}}{s^2}$$

- **Second approach:** exploiting the **identity theorem for polynomials**.

$$C_{1,1}s^2(s+2) + C_{2,1}s^2(s+1) + C_{3,1}s(s+1)(s+2) + C_{3,2}(s+1)(s+2) = s+3$$

$$s=0 \longrightarrow 2C_{3,2} = 3 \longrightarrow C_{3,2} = \frac{3}{2} \qquad s=-1 \longrightarrow C_{1,1} = 2$$

$$s=-2 \longrightarrow -4C_{2,1} = 1 \longrightarrow C_{2,1} = -\frac{1}{4}$$

$$s=+1 \longrightarrow 3C_{1,1} + 2C_{2,1} + 6C_{3,1} + 6C_{3,2} = 4 \longrightarrow C_{3,1} = -\frac{7}{4}$$

Example: Solution of a Linear Differential Equation (cont.)

$$Y_f = \frac{2}{(s+1)} - \frac{1}{4} \frac{1}{(s+2)} - \frac{7}{4s} + \frac{3}{2s^2}$$

$$\mathcal{L}(Y_f(s)) = y_f(t) = \left[\frac{3}{2}t - \frac{7}{4} + 2e^{-t} - \frac{1}{4}e^{-2t} \right] \cdot 1(t)$$

- The solution of the ODE equation is:

$$\underline{y(t) = y_l(t) + y_f(t)}$$

The Laplace Transform: Initial and Final Value Theorems

Can we determine the initial value of a time-domain function from its Laplace transform?

Can we determine the steady-state value of a time-domain function from its Laplace transform?

Initial Value Theorem

- Given a Laplace transform $F(s)$ that is a **strictly proper rational function**, the following property holds

$$\lim_{t \rightarrow 0} f(t) = \lim_{s \rightarrow \infty} s F(s)$$

This is a useful theorem: one can immediately analyse the initial behaviour of $f(t)$ directly from its Laplace transform $F(s)$!

Final Value Theorem

- If all the **poles** of $F(s)$ are in the **left open half-plane**, with **at most one simple pole in the origin**, then **the two limits both exist finitely** and **assume equal values**.

$$\lim_{t \rightarrow +\infty} f(t) = \lim_{s \rightarrow 0} s \cdot F(s)$$

This is a useful theorem: one can immediately find the steady-state value of $f(t)$ directly from its Laplace transform $F(s)$!

Application Example

- Consider the Laplace transform $F(s)$:

$$F(s) = \frac{5s + 3}{(s + 1)(s + 2)^2}$$

- Determine

$$f(0) = ?$$

$$\lim_{t \rightarrow \infty} f(t) = ?$$

$$\dot{f}(0) = ?$$